

CHAPTER 11

HUMAN IMPACTS ON THE SEA

CHAPTER OUTLINE AND SIGNIFICANT ELEMENTS

“It was the best of times, it was the worst of times, it was the age of wisdom, it was the age of foolishness, it was the epoch of belief, it was the epoch of incredulity, it was the season of light, it was the season of darkness, it was the spring of hope, it was the winter of despair. . . .”

Charles Dickens, 1859

A Tale of Two Cities

We humans use the oceans as our larder, sewer, and trash barrel. In the future, we may use it as a source of new wonder drugs. Each of these functions is a type of ecosystem service. We also view the oceans with a sense of awe and wonder, and potentially with a sense of doom if global warming results in melting of the polar ice caps. These topics are explored in seven sections: FISHERIES, MARICULTURE, POLLUTION, MARINE DISEASES, DRUGS FROM THE SEA, GLOBAL WARMING AND SEA-LEVEL CHANGE, and CONCLUDING REMARKS.

Ecosystem Services

“Ecosystem Services are the processes by which the environment produces resources that we often take for granted such as clean water, timber, and habitat for fisheries, and pollination of native and agricultural plants. Whether we find ourselves in the city or a rural area, the ecosystems in which humans live produce good and services that are very familiar to us.”

Ecological Society of America, <http://www.esa.org/education/edupdfs/ecosystems-services.pdf>

Ecosystem services can be overexploited and degraded to a point where the use (abuse) of one ecosystem service by a stakeholder leads to the loss of another ecosystem service important to some other stakeholder(s). These conflicts can be manifest over space and time. For example, the use of a wetland as a landfill for a city's trash may lead to groundwater contamination that is not seen for a decade or more. Pollution from various sources also may reduce biodiversity of tropical reefs; this loss eliminates potential resources for new drugs. For many centuries, we humans have been conducting our affairs with the notion that our activities are composed solely of internal factors, or internalities; these internalities can and must be controlled for the activities to flourish. However, our activities often generate by-products with which we then must deal. Examples include sewage, excess carbon dioxide, and erosion. When these by-products can be discarded with little or no cost to the overall activity, they are termed externalities (that is, their real costs have been externalized). We need to learn how to internalize these externalities while still carrying on our business efficiently.

Alcamo, J., and E.M. Bennett. 2003. *Ecosystems and human well-being: a framework for assessment*. Island Press: Washington, DC.

Costanza, R., et al. 1997. The value of the world's ecosystem services and natural capital. *Nature* 387: 253–260.

Daily, G.C. (ed.). 1997. *Nature's services: societal dependence on natural ecosystems*. Island Press: Washington, DC.

Ehler, C.N., and D.J. Basta. 1993. Integrated management of coastal areas and marine sanctuaries: a new paradigm. *Oceanus* 36(3): 6–13.

Fonseca, M.S., B.E. Julius, and W.J. Kenworthy. 2000. Integrating biology and economics in seagrass restoration: How much is enough and why? *Ecol. Eng.* 15: 227–238.

Heal, G.M. 2000. *Nature and the marketplace: capturing the value of ecosystem services*. Island Press: Washington, DC.

Holmlund, C.M., and M. Hammer. 1999. Ecosystem services generated by fish populations. *Ecol. Econ.* 29: 253–268.

- Jenkins, M. 2003. Prospects for biodiversity. *Science* 302: 1175–1177.
- Loucks, O.L., and R.F. Gorman. 2004. Regional ecosystem services and the rating of investment opportunities. *Front. Eco. Env.* 4: 207–216.
- Moberg, F., and P. Ronnback. 2003. Ecosystem services of the tropical seascape: interactions, substitutions and restoration. *Ocean Coast. Manag.* 46: 27–46.
- Rudd, M.A., et al. 2003. Policy analysis for tropical marine reserves: challenges and directions. *Fish Fisheries* 4: 65–86.
- Tilman, D. 1999. Global environmental impacts of agricultural expansion: the need for sustainable and efficient practices. *PNAS* 96: 5995–6100.
- Yap, H.T. 2000. The case for restoration of tropical coastal ecosystems. *Ocean Coast. Manag.* 43: 841–851.

I. Fisheries

The first part of this chapter is separated into nine sections:

- Major Fishing Areas
- Major Commercial Species
- Sustainable Yield and the Future
- Overexploitation
- Ecosystem Changes
- Solutions
- Tragedy of the commons
- Regulation
- New Fisheries

Science

algin	fisheries	mollusks
anchoveta	fishing gear	overexploitation
baleen whales	flatfishes	overfishing
blue whales	gadoids	Peruvian anchoveta
bycatch	Garrett Hardin	pollock
California gray whale	ghost fishing	ROV
California sardine	haddock	sardine
<i>Cannery Row</i>	hake	sharks
clupeoid	harpoon gun	shell mounds
Cod War	herring	shellfish
cods	John Steinbeck	sustainable yield
demersal fish	krill	target fish
El Niño	mackerel	tragedy of the commons
<i>Engraulis ringens</i>	marine mammals	whales
<i>Euphausia superba</i>	maximum sustainable yield	
euphausiids	middens	

Geopolitics

CCOFO	International Halibut Commission (IHC)
Exclusive Economic Zone (EEC)	International North Pacific Fisheries Commission (INPFC)
Fur Seal Treaty	International Tuna Commission (ITC)
International Commission Conservation Atlantic Tuna (ICCAT)	International Whaling Commission (IWC)
	Law of the Sea

1. The archaeological evidence from middens (dump heaps) indicates that humans have had a fondness for shellfish for a very long time. Tapping the resources of your campus archaeologist to give a guest lecture on this topic might be worth a try.
2. Although the continental shelves are by far the richest fisheries, they are uneven in their production of commercially important fish. The most important fish include herrings, sardines, and anchovies; these fish are clupeoids that feed on the first or second trophic level. However, most of this harvest is not consumed directly by humans. Instead, it is processed as protein meal and used in animal foods.
3. The second most important fish harvested include cods, haddock, pollock, and hake. These gadoids are bottom dwellers, or demersal fish. Also harvested are scombroids (swiftly swimming fish commonly known as mackerel). Tuna, of course, are very important as an open-ocean fish. Many other fish also are important, including redfish, bass, flatfishes, and sharks.
4. A wide variety of shellfish constitute a sizable fishery, including various crabs, shrimps, lobsters, and various mollusks. In a number of countries, more “exotic” foods are harvested, including sea cucumbers, polychaete worms, sea urchins (for their gonads), and jellyfish.
5. The problem with the world’s fisheries is that they are not an inexhaustible supply of protein for a human population in exponential growth. The trick is to determine the maximum sustainable yield of seafood and then keep exploitation to within that limit. The decline in recent years of certain fish stocks indicates that we have been engaged in overexploitation of this resource. However, not all of these declines are solely a matter of mismanagement (really nonmanagement); some are related to large-scale fluctuation in climate and water currents.

Baker, B. 2000. Fisheries management faces many challenges. *BioScience* 50: 652.

Cheong, S-M. 2004. Managing fishing at the local level: the role of fishing village cooperatives in Korea. *Coastal Manag.* 32: 191–201.

Constance, D.H., and A. Bonanno. 1999. Contested terrain of the global fisheries: “dolphin-safe” tuna, the Panama Declaration, and the Marine Stewardship Council. *Rural Soc.* 64(4): 597–623.

Hall, S.J., and B. Mainprize. 2004. Towards ecosystem-based fisheries management. *Fish Fisheries* 5: 1–20.

Hastings, A., and L.W. Botsford. 1999. Equivalence in yield from marine reserves and traditional fisheries management. *Science* 284: 1537–1538.

Jaffa, M.R. 1997. There aren’t plenty more fish in the sea. *Nature* 387: 339.

Koslow, J.A. 1997. Seamounts and the ecology of deep-sea fisheries. *Am. Sci.* 85: 168–176.

Nicol, S., and W. de la Mare. 1993. Ecosystem management and the Antarctic krill. *Am. Sci.* 81(1): 36–47.

Normile, D. 2000. Japan’s whaling program carries heavy baggage. *Science* 289: 2264–2265.

Parfit, M. 1995. Diminishing returns. *Nat. Geog.* 188(5): 2–37.

Pauli, D., et al. 2000. Fishing down aquatic food webs. *Am. Sci.* 88: 46–51.

Reid, T.R. 1995. The great Tokyo fish market. *Nat. Geog.* 188(5): 40–55.

Safina, C. 1995. The world’s imperiled fish. *Sci. Am.* 273(5): 46–53.

Satria, A., and Y. Matsida. 2004. Decentralization policy: an opportunity for strengthening fisheries management system? *J. Environ. Develop.* 13: 179–196.

Williams, N. 1998. Overfishing disrupts entire ecosystems. *Science* 279: 809.

6. One example of a fish that has incurred lower yields due to changes in climate and current flow is the Peruvian anchoveta, *Engraulis ringens*. Students should be reminded of the effects of water conditions (El Niño and the Southern Oscillation [ENSO]) on trophic-level production.

7. I have spoken to professors of English at various institutions who insist that the character Doc in John Steinbeck's novel *Cannery Row* is Steinbeck himself. Most marine biologists with whom I have spoken insist that Doc more closely resembles Ed Ricketts. The friendship between Ricketts and Steinbeck is explored at various levels in the following references:

Astro, R. 1973. *John Steinbeck and Edward F. Ricketts: the shaping of a novelist*. Minneapolis: University of Minnesota Press.

Hedgpeth, J.W. 1978a. *The Outer Shores, Part 1: Ed Ricketts and John Steinbeck explore the Pacific coast*. Eureka, CA: Mad River Press, Inc.

Hedgpeth, J.W. 1978b. *The outer shores, Part 2: Breaking through*. Eureka, CA: Mad River Press, Inc.

Ricketts, E.F., et al. 1985. *Between pacific tides*. 5th ed. Stanford, CA: Stanford University Press.

Steinbeck, J., and E.F. Ricketts. 1941. *Sea of Cortez: a leisurely journal of travel and research*. New York: Viking Press.

8. The efforts of the International Whaling Commission (IWC) begun in 1948 have met with less than modest success. Its charge was to analyze data on this fishery and to set quotas to conserve the species. Lacking enforcement abilities, the IWC is often ignored. Many even believe that the IWC is not an agency attempting to conserve the whale stock, but rather a cheerleading squad for the status quo.

9. Garrett Hardin's *The Tragedy of the Commons* is an essay worthy of devoting time for classroom discussion. A panel discussion or debate may best serve large classes in which too few students would otherwise be able to participate.

Anderson, E.N., and B.R. Warren. 1998. Reevaluating the tragedy of the commons. *Cons. Bio.* 12: 1168–1169.

Hardin, G. 1968. The tragedy of the commons. *Science* 162: 1243–1248.

Holden, C. 2003. *Tragedy of the Commons* author dies. *Science* 302: 32.

Kennedy, D. 2003. Sustainability and the commons. *Science* 302: 1861. [Editorial on Garrett Hardin's essay, "Tragedy of the Commons."]

10. Some fish are currently being considered for greater exploitation, including krill, squid, and under-utilized fish species.

II. Mariculture

This part of the chapter is separated into four sections:

- History and Extent
- Species Cultivated
- Yields
- Problems and Restrictions to Mariculture Development

aquaculture
crustaceans

fishes
mariculture

mollusks
yield

Aquaculture, or, as it is referred to in the marine setting, mariculture, attempts to cultivate aquatic organisms under controlled conditions for human and animal consumption and for certain products (such as pearls). So far, few organisms are cultivated in this way; such organisms include shrimps, oysters, mussels, salmon, and various other fishes. While the current production rates are rather low, there is considerable potential for production of valuable species in mariculture. Countries that have established mariculture programs include China, India, Indonesia, Israel, Japan, Spain, and Taiwan.

III. Pollution

The solution to pollution is dilution.
—rhyme of the ancient sewage engineer

This part of the chapter is separated into eight sections:

- Oil
- Sewage and Garbage
- Chemicals
- Radioactive Wastes
- Thermal Pollution
- Eutrophication
- Biological Invasions
- Miscellaneous Human Impacts

alien (exotic) animals	<i>Exxon Valdez</i>	oil seep
alien biota	FDA	Plutonium 238/239
<i>Amoco Cadiz</i>	foams	radionuclides
biological magnification	garbage pollution	red tide
Coal Oil Point	half-life	remobilization
crude oil	high-level wastes	sewage
Ctenophora	ionization radiation	slicks
curies	Ixtoc I well	slimes
DDD	Kaneohe Bay	subseabed emplacement
DDE	lateral transport	thermal pollution
DDT	low-level wastes	<i>Torrey Canyon</i>
dead zones	methyl mercury	uptake
deep-sea disposal	Minamata Bay	vertical diffusion
<i>Dictyosphaeria</i>	Minamata disease	White's Point
dimethyl sulfide (DMS)	<i>Mnemiopsis</i>	
Dutch elm disease	NOAA-CNEXO	
eggshell thinning	ocean dumping	
eutrophication	offal	

1. Oil, sewage, garbage, nuclear and chemical wastes, and excess heat, the by-products of our modern industrialized society, all end up in the world's oceans. The release of untreated or improperly treated sewage still causes problems in aquatic systems. Oil spills are very dramatic examples of our mistreatment of the marine habitat, and, of course, they make for excellent headlines for TV and newspapers. The text details several of the major oil spills of the past few decades.
2. While major oil spills make headline news, they are fairly rare events. On the other hand, use of the oceans as large cesspools for our waste products is a chronic and widespread problem of incomprehensible magnitude. The expediency of dumping these wastes so cheaply is that they are immediately out of sight and therefore out of mind. But, as indicated in the examples reviewed in this text, this dumping has dramatic consequences.

Anon. 1993. Beaches awash in plastic. *Science News* 144: 235.

Barkdull, J. 1998. Nixon and the marine environment. *Presidential Studies Quarterly* 28(3): 587.

Benton, T.G. 1995. From castaways to throwaways: marine litter in the Pitcairn Islands. *Biol. J. Linnean Soc.* 56: 415–422.

Burger, J. (ed.). 1994. *Before and after an oil spill*. New Brunswick, NJ: Rutgers University Press.

Grove, N. 1978. Black day for Brittany. *Nat. Geo.* 154(1): 124–135.

Grove, N. 1978. Sailing with the supertankers. *Nat. Geo.* 154(1): 102–124.

Hodgson, B. 1990. Alaska's big spill—can the wilderness heal? *Nat. Geo.* 177(1): 5–43.

- Johnson, D.C., et al. 1995. Detection of giardia and cryptosporidium in marine waters. *Water Sci. Techn.* 31: 439–442.
- Lee, D.B. 1989. Tragedy in Alaska waters. *Nat. Geo.* 176(2): 260–263.
- Mallin, M.A. 2000. Impacts of industrial animal production on rivers and estuaries. *Am. Sci.* 88: 26–37.
- Palmisano, A.C., and C.A. Pettigrew. 1992. Biodegradability of plastics. *BioScience* 42(9): 680–685.
- Reish, D.J., P.S. Oshida, and A.J. Mearns. 1994. Effects of pollution on saltwater organisms. *Water Environ. Res.* 66: 623–635.
- Reish, D.J., P.S. Oshida, and A.J. Mearns. 1997. Effects of pollution on saltwater organisms. *Water Environmental Res.* 69: 877–892.
- Reish, D.J., P.S. Oshida, and A.J. Mearns. 1998. Effects of pollution on saltwater organisms. *Water Environmental Res.* 70: 931–949.
- Ryan, P.G., and C.L. Moloney. 1993. Marine litter keeps increasing. *Nature* 361: 23.
3. The places of abuse reviewed in this chapter are White's Point off Palos Verdes Peninsula, CA (sewage), Kaneohe Bay, HI (sewage), New York City (sewage, among others), and Minamata Bay, Japan (industrial wastes). This last tragedy spawned a neurological disorder and severe birth defects collectively referred to as Minamata disease.
- The Minamata tragedy (ca. 1953–1960) occurred for two reasons. First, a factory that was producing plastics had been releasing Hg waste into Minamata Bay. Second, bacteria are capable of methylating mercury, probably through a biochemical reaction known as transmethylation. The resulting mono- and dimethyl mercury compounds are lipid soluble and can enter the food web from the lowest trophic levels. They are then concentrated up the food chain by biomagnification into the fish. Fish caught from these waters had high concentrations of Hg in their tissues ($\approx 30\text{--}100\text{ mg/Kg}$ dry weight). More than 100 people were affected, and just under 50% of those affected died. A number of children who were born to the women of Minamata Bay suffered from gross birth defects.
- Allchin, D. 1999. The tragedy and triumph of Minamata. *The American Biology Teacher* 61: 413–419.
- Cavanagh, J.B. 2000. The pathology of Minamata disease. *Brain* 123: 195–196.
- Futatsuka, M., et al. 2000. Health Surveillance in the Population Living in a Methyl Mercury-Polluted Area over a Long Period. *Env. Res.* 83: 83–92.
- Tsubaki, T., and K. Irukayama. 1977. *Minamata disease: methylmercury poisoning in Minamata and Niigata, Japan*. Tokyo: Kodansha Scientific Ltd.
- Watts, J. 1997. Minamata Bay finally declared free of mercury. *Lancet* (North American edition) 350: 422.
4. Both low- (10^{-8} curies per gram) and high-level ($> 10^{-8}$ curies/g) radioactive wastes are (or have been) dumped at sea. A good rule of thumb is to isolate radioactive materials for a time equal to at least $15\times$ their half-life. For some of the most dangerous radioactive materials, this length of time can be quite long. Plutonium 239 has a half-life of 24,100 years!
- Deere-Jones, T. 1991. Back to the land: the sea-to-land transfer of radioactive pollution. *The Ecologist* 21(1): 18–23.
- Yablokov, A.V. 2001. Radioactive waste disposal in seas adjacent to the territory of the Russian Federation. *Marine Poll. Bull.* 43: 8–17.
5. Silt from dredging, thermal pollution, eutrophication, specific chemicals (such as DDT and its decomposition products, DDD and DDE), Dimethyl Sulfide (DMS) and the transportation of exotic organisms to new habitats are other problems with which we must contend. All of these abuses are anthropogenic.
- Burke, M. 1994. Phosphorous fingered as coral killer. *Science* 263: 1086.
- Carlton, J.T., et al. 1990. Remarkable invasion of San Francisco Bay (California, USA) by the Asian clam *Potamocorbula amurensis*: I. Introduction and dispersal. *Marine Ecol. Prog. Ser.* 66: 81–94.

Carson, R. 1962. *Silent spring*. New York: Fawcett Crest Books.

Claudi, R., and G.L. Mackie 1993. *Practical manual for zebra mussel monitoring and control*. Boca Raton, FL: Lewis Publishers.

International Maritime Organization. International Guidelines for Preventing the Introduction of Unwanted Aquatic Organisms and Pathogens from Ships' Ballast Water and Sediment Discharges. Resolution MEPC.50(31), as adopted 4 July 1991. MEPC 31/21, Annex 16. 16pp.

Kennish, M.J. 1992. *Ecology of estuaries: anthropogenic effects*. Boca Raton, FL: CRC Press.

Kennish, M.J. (ed.). 1997. *Practical handbook of estuarine and marine pollution*. Boca Raton, FL: CRC Press.

Lipske, M. 2000. How Rachel Carson helped save the brown pelican. *Nat. Wildl.* 38(1): 44–50.

Muir, D., F. Riget, and M. Cleemann. 2000. Circumpolar trends of PCBs and organochlorine pesticides in the Arctic marine environment inferred from levels in ringed seals. *Environ. Sci. Tech.* 34(12): 2431–2438.

Nalepa, T.F., and D.W. Schloesser. 1993. *Zebra mussels: biology, impacts, and control*. Boca Raton, FL: Lewis Publishers.

Roberts, L. 1990. Zebra mussel invasion threatens U.S. waters. *Science* 249: 1371–1372.

Rosta, P. 2000. Deepwater cap to cover DDT off southern California coast. *ENR* 224(16): 22.

Smithsonian's The Ocean Planet. "Aliens Among Us." http://seawifs.gsfc.nasa.gov/OCEAN_PLANET/HTML/ps_alien.html. Active as of June 17, 2004.

IV. Marine Diseases

<i>Acropora</i>	ENSO	pathogen
bacteria	gray seals	<i>Perkinsus</i>
coral	<i>Halichorus gryphus</i>	<i>Pfisteria piscicida</i>
<i>Crassostrea virginica</i>	harbor seals	<i>Phoca vitulina</i>
dermo	host shift	protozoan
<i>Diadema antillarum</i>	morbillivirus	sea urchins
dinoflagellates	oyster	<i>Vibrio cholerae</i>

A disease is not just a "thing," whether in a human or in any other organism. Disease is a metabolic upset in the host organism as caused by some pathogen, such as bacteria, a fungus, a protozoan, or a virus. See the Enrichment Exercise in Chapter 9 on Koch's postulates.

Burkholder, J., and J. Kaiser. 2004. Ongoing controversy over Pfiesteria. *Science* 304: 46–47. [letter to the editor]

Harvell, C.D., et al. 1999. Emerging marine diseases—climate links and anthropogenic factors. *Science* 285: 1505–1510.

Smith, G.W., et al. 1996. Caribbean sea-fan mortalities. *Nature* 383: 487.

V. Drugs from the Sea

antitumor	marine pharmacology	sponges
cnidarians	palytoxin	squalamine
ichthyotoxin	saponin	tunicates

Researchers have only begun to explore the vast resources of marine organisms that may hold within their tissues unique compounds useful in the treatment of disease. Although a detailed exploration of this topic is beyond the scope of an introductory textbook, a few examples are offered to tantalize us.

Alonso, D., et al. 2003. Drugs from the sea: conotoxins as drug leads for neuropathic pain and other neurological conditions. *Mini Rev. Medicinal Chem.* 3: 785–787.

Cohen, T. 1993. Pharmaceuticals from the sea. *Tech. Rev.* 96(3): 15–16.

VI. Global Warming and Changes in Sea Level

For a different perspective on doom theories, read Robert Frost's poem *Fire and Ice*.

carbon dioxide (CO ₂)	greenhouse effect	limiting factor
chlorofluorocarbons (CFCs)	greenhouse gases	methane (CH ₄)
clear cutting	ice melting	Moss Landing Marine Lab
CO ₂ sink	iron fertilization	nitrous oxide
global warming	IronEx experiments	sea-level rise

Doom theories are never much fun. This section briefly reviews global warming. Unfortunately, the predictions from this theory seem all too accurate. Another problem related to global warming is depletion of the ozone layer. Following are several readings and Web sites that might get your students started on these issues of environmental abuse:

- Bourne, J. 2000. Louisiana's vanishing wetlands: going, going. . . . *Science* 289: 1860–1963.
- Ellis, W.S. 1990. A Soviet sea lies dying. *Nat. Geo.* 177(2): 73–93.
- Gehrels, W.R., et al. 2002. Rapid sea-level rise in the Gulf of Maine, USA, since A.D. 1800. *The Holocene* 12: 3833–3839.
- Johannessen, O.M., E.V. Shalina, and M.W. Miles. 1999. Satellite evidence for an Arctic sea ice cover in transformation. *Science* 286: 1937–1939.
- Kerr, R.A. 2000. Ice, mud point to CO₂ role in glacial cycle. *Science* 289: 1868.
- Lee, D.W., D.K. Pak, and H.J. Spero. 2000. Climate impact of late Quaternary Equatorial Pacific sea surface temperature variations. *Science* 289: 1719–1724.
- Nerem, R.S. 1995. Global mean sea level variations from TOPEX/POSEIDON altimeter data. *Science* 268: 708–710.
- Roemmich, D. 1992. Ocean warming and the sea level rise along the southwest U.S. coast. *Science* 257: 373–375.
- Rogers, D.J., and S.E. Randolph. 2000. The global spread of malaria in a future, warmer world. *Science* 289: 1763–1766.
- Shackleton, N.J. 2000. The 100,000-year ice-age cycle identified and found to lag temperature, carbon dioxide, and orbital eccentricity. *Science* 289: 1897–1902.
- Taylor, K. 1999. Rapid climate change. *Am. Sci.* 87: 320–327.
- Vinnikov, K.Y., et al. 1999. Global warming and northern hemisphere sea ice extent. *Science* 286: 1934–1937.
- Yohe, G.W., and M.E. Schlesinger. 1998. Sea-level change: the expected economic cost of protection or abandonment in the United States. *Climatic Change* 38: 447–472.

VII. Concluding Remarks

Will the oceans become the next environmental tragedy? Some believe that they already have. What do your students think, and what sort of evidence can they marshal to argue their position? A debate in which both sides are represented would be an interesting idea.

ENRICHMENT EXERCISES

Environmental Ethics

I suspect that some instructors will wish to wrap up their course with a discussion on ethics, perhaps using Hardin's essay as a starting point. The following references and Web sites are selected to provide ideas for initiating such a discourse:

- Allen, W.H. 1992. Increased dangers to Caribbean marine ecosystems. *BioScience* 42(5): 330–335.
- Allott, P. 1992. Mare nostrum: a new international law of the sea. *Am. J. Internat. Law* 86(4): 764–787.

- Baker, C.S., and S.R. Palumbi. 1994. Which whales are hunted? A molecular genetic approach to monitoring whaling. *Science* 265: 1538–1539.
- Béland, P. 1996. The beluga whales of the St. Lawrence River. *Sci. Am.* 274(5): 74–81.
- Birnie, P.W. 1992. An EC exclusive economic zone: marine environmental aspects. *Ocean Dev. Internat. Law* 23(2–3): 193–216.
- Cleveland, H. 1993. The global commons. *The Futurist* 27(3): 913.
- Cohen, J.E. 1995. *How many people can the earth support?* New York: W.W. Norton and Company.
- Edmondson, W.T. 1991. *The uses of ecology: Lake Washington and beyond.* Seattle: University of Washington Press.
- Gore, A. 1992. *Earth in the balance.* Boston: Houghton Mifflin.
- Hardin, G. 1968. The tragedy of the commons. *Science* 162: 1243–1248.
- Lamb, J., and P. Gizewski. 1993. The policy makers' challenge: radioactive dumping in the Arctic Ocean. *Oceanus* 36(3): 89–91.
- Macdonald, J.M. 1994. Artificial reef debate: habitat enhancement or waste disposal? *Ocean Dev. Internat. Law* 25(1): 87–118.
- Millman, J.D. 1992. *Climate change and the level of seas.* London: Edward Arnold.
- Moore, S.J. 1992. Troubles in the high seas: a new era in the regulation of U.S. ocean dumping. *Environ. Law* 22(3): 913–951.
- Oerlemans, J. 1994. Quantifying global warming from the retreat of glaciers. *Science* 264: 243–245.
- Revelle, R. 1982. Carbon dioxide and world climate. *Sci. Am.* 247(2): 35–43.
- Ryan, P.G., and C.L. Moloney. 1993. Marine litter keeps increasing. *Nature* 361: 23.
- Sahagian, D.L., F.W. Schwartz, and D.K. Jacobs. 1994. Direct anthropogenic contributions to sea level rise in the twentieth century. *Nature* 367: 54–57.
- Schachte, Jr., W.L. 1992. The value of the 1982 UN convention on the Law of the Sea: preserving our freedoms and protecting the environment. *Ocean Dev. Internat. Law* 23(1): 55–69.
- Schneider, S.H. 1994. Detecting climatic change signals: are there any “fingerprints”? *Science* 263: 341–347.
- Shaver, G.R., et al. 1992. Global change and the carbon balance of Arctic ecosystems. *BioScience* 42(6): 433–441.
- Skerrett, P.J. 1992. Nuclear burial at sea. *Tech. Rev.* 95(2): 22–23.
- Stevenson, J.R., and B.H. Oram. 1994. The future of the United Nations Convention on the Law of the Sea. *Am. J. Internat. Law* 88(3): 488–499.

Plastics and Trash in the Marine Environment

What constitutes the trash on our beaches? Is it mostly plastics, paper, or glass? Is it a combination of all three? Does it consist of other materials as well? Depending on the level of ecological studies your students have completed, you might have them compare the species richness or species diversity of trash in two or more marine habitats. A comparison of rank-abundance curves also might be instructive. For example, consider garbage found on the oceanic and inland sides of a barrier island or a sandy beach versus that found in a calm estuary. What are the sources of all this trash? Do beaches differ in the species composition of trash? Having your students clean up an area while at the same time analyzing the composition of the beach trash can serve a variety of purposes: It is good PR for your school; it is good physical exercise for the students; it is good for the students to see some of the same stuff they toss out turn up on the beach (this aspect of the exercise promotes self-reflection); it is a good way to get the students involved in environmental matters; it is a good way to initiate discussions on a variety of topics. Questions that can be explored include the following: Who dumps the stuff? How long does the trash last in the environment? What is the impact of the garbage on the environment?

Please caution your students to be very careful about what they pick up, how they pick it up, and to what lengths they will go to pick it up. Having the students wear thick gloves is an excellent idea.

